

Publication: Functional Connectivity Predicts Working Memory Performance -- A Multi-Site fMRI Study

Authors

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Abstract

This multi-site fMRI study examined whether functional connectivity between prefrontal and parietal regions predicts working memory performance.

Methods

Participants: A total of 120 participants were recruited from two sites. After quality control, 98 participants were included in the final analysis.

MRI Acquisition:

Preprocessing: fMRIPrep 23.1.0. No additional denoising was applied.

Primary Outcome: DLPFC-PPC functional connectivity (Fisher z-transformed) correlated with N-back accuracy.

Analysis: Mixed-effects regression with working memory accuracy as dependent variable, connectivity + age + sex as independent variables, site as covariate. Software: FSL 6.0 + Python Nilearn. No permutation testing was performed; standard p-values with FDR correction were used.

Results

Functional connectivity between DLPFC and PPC significantly predicted working memory accuracy ($r=0.34$, $p=0.002$). The effect was consistent across both scanner sites.

Conclusions

Functional connectivity is a robust biomarker of working memory that generalizes across scanner manufacturers.